# Bragg band gaps in a LEGO sonic crystal along $\Gamma X$ and $\Gamma M$

Afonso Duarte (ist1118524) · Filipe Marcelino (ist1117896) · Inês Ferreira (ist1117578) · José Santos (ist1118707)
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**Abstract.** A sonic crystal blocks sound in a band of frequencies, the band gap, expected near $f_B = c/(2d)$, with $d$ the spacing between planes of scatterers. We measured the transmission of white noise through square lattices of $2 \times 2$ LEGO columns, with lattice constant $a = 32$–$56$ mm and $N = 1$–$11$ rows, along $\Gamma X$ and $\Gamma M$ (lattice rotated by 45°), and compared it with a plane-wave expansion (PWE). Along $\Gamma X$ the lower gap edge agrees with the PWE within $1.3\sigma$ for $a \leq 48$ mm but not for $a = 56$ mm ($3.9\sigma$); the gap does not scale as $1/a$ ($\chi^2/\nu = 15$), and the attenuation grows linearly with $N$, by $1.45 \pm 0.29$ to $1.98 \pm 0.41$ dB per row ($a = 40$–$56$ mm). Rotating the crystal to $\Gamma M$ raises the lower edge by a factor 1.27–1.45, consistent with $\sqrt{2}$ within $2\sigma$. The precision is limited by the reference measurement and by sound bypassing the crystal, and we find no evidence of a complete gap.

## 1 Introduction

A sonic crystal is a periodic array of rigid obstacles in air. Reflections from successive planes of obstacles, separated by $d$, add in phase when $\lambda = 2d$, and around

$$f_B = \frac{c}{2d}, \qquad d = a \ (\Gamma X), \qquad d = a/\sqrt{2} \ (\Gamma M) \tag{1}$$

a band gap opens [1, 2]: at the edge of the Brillouin zone, $k = \pi/d$, the standing waves with maximum pressure on the planes of bricks and in the air between them no longer have the same frequency, and between the two, $f_{\text{inf}} < f_B < f_{\text{sup}}$, no wave propagates. Inside the gap $k = \pi/d + i\kappa$ and the pressure decays as $e^{-\kappa x}$; after $N$ rows the attenuation in decibels is

$$D(N) = \alpha N, \qquad \alpha = 20 \log_{10}(e)\, \kappa d. \tag{2}$$

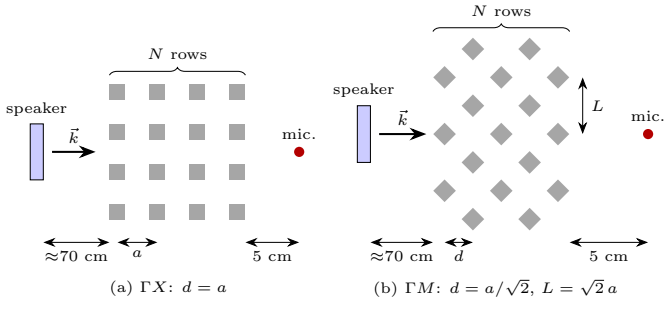
The position and width of the gap depend on the filling fraction $ff = s^2/a^2$ (0.24 to 0.08 here): the bricks slow the sound down, $c_{\text{eff}} \simeq c/\sqrt{1+ff}$ [3], and the gap moves below $f_B$, more so for smaller $a$. The bands are computed by solving $\nabla \cdot (\rho^{-1} \nabla p) + \omega^2 p/(\rho c^2) = 0$ with $p$ written as a Bloch wave (PWE [4], Appendix B). We tested, along $\Gamma X$ and $\Gamma M$, the gap edges, their dependence on $a$, the linear attenuation in $N$, and whether the gaps of the two directions overlap (complete gap).

## 2 Method

***Experimental set-up.*** The crystal is a square lattice of columns (Fig. 1), each one a stack of $2 \times 2$ LEGO bricks (side $s = 15.8$ mm) on a LEGO baseplate. The lattice constant $a$ is the centre-to-centre distance between neighbouring columns.

Along $\Gamma X$ the rows are at $d = a$. Only this spacing along the propagation direction sets the Bragg frequency, $f_B = c/(2d)$, and it was the only one changed: the lateral spacing and the enclosure were kept fixed. Along $\Gamma M$ the lattice is rotated by 45° and the two spacings can no longer be chosen separately: the planes are at $d = a/\sqrt{2}$ and the columns within a plane at $L = \sqrt{2}\, a = 2d$, so $f_B = c/L$. Increasing $a$ therefore widens the lattice laterally as well.

The crystal is enclosed by LEGO walls and roof and placed in a cardboard box with foam around it. A Bluetooth loudspeaker $\approx 70$ cm from the crystal face emits white noise, with frequencies between 500 and 12 000 Hz, for 10 s, and a USB microphone 5 cm behind the crystal records the sound. The loudspeaker distance satisfies the plane-wave criterion (Appendix B).




(a) $\Gamma X$: $d = a$ (b) $\Gamma M$: $d = a/\sqrt{2}$, $L = \sqrt{2}\, a$

**Figure 1.** Set-up seen from above (not to scale). (a) $\Gamma X$: aligned columns. (b) $\Gamma M$: lattice rotated by 45°, with a brick vertex facing the loudspeaker and rows alternating between 4 and 3 bricks. $a$: distance between columns; $d$: distance between planes; $L$: period within a plane.

***Parameters varied and repetitions.*** We varied $a = 32$, 40, 48 and 56 mm along $\Gamma X$ and $a = 32$, 40 and 48 mm along $\Gamma M$, and the number of rows $N$, removing one row at a time from the microphone side ($N = 10 \to 1$ along $\Gamma X$ and $N = 11 \to 1$ along $\Gamma M$). Each configuration had 5 recordings, for a total of 72 configurations and 360 recordings, of which 16 were excluded for saturation or Bluetooth dropouts. The 4 sessions took place at $\theta = 20.5\,^\circ$C and $20.0\,^\circ$C ($\pm 1\,^\circ$C), which gives $c = (343.5 \pm 0.6)$ m/s and $(343.2 \pm 0.6)$ m/s.

***Quantity measured and data processing.*** We measured the relative transmission

$$T(f) = 10 \log_{10} \left[ \frac{\bar{P}_{\text{crystal}}(f)}{\bar{P}_{\text{ref}}(f)} \right] \ (\text{dB}), \tag{3}$$

where $P$ is the power spectral density of the recorded signal, and the reference is the same set-up without columns. The division removes the response of the loudspeaker, the microphone and the room.

*Spectra.* $P$ was computed with Welch's method [5] (Hann window, 4096 points, $\Delta f = 11.7$ Hz at a sampling rate of 48 kHz) on the segment 0.8–10.3 s of each recording, which excludes the Bluetooth delay.

*Smoothing.* To reduce the statistical fluctuations of the spectrum, which are amplified when dividing two spectra, each $P$ was smoothed in power. The window is 1/6 of an octave wide (±6 % of $f$).

*Reference.* The reference was measured for each value of $a$ at the start of the measurements. For each $a$ we used, among the references of the session, the one that makes $T(f)$ flattest in the control band (see below); the others enter the uncertainty associated with the reference.

***Control band.*** As a check of the measurement, $T$ should be constant and close to 0 dB in the band 0.35–0.70 $f_0$, with $f_0 = c/(2a)$ along both directions. This band lies at frequencies below the first gap of the lattice, where the crystal is practically transparent.

## 3 Analysis

From each $T(f)$ we extract $T_0$ and the ripple (median and robust deviation of $T$ in the control band, Sec. 2); the gap edges $f_{\text{inf}}$ and $f_{\text{sup}}$, where $T$ crosses $T_0 - 6$ dB around 0.75–1.25 $f_B$; and the depth $D = T_0 - \langle T \rangle_{f_B \pm 10\%}$. For each $a$, the edges are the weighted mean over $N \geq 4$; these are compared with the PWE and fitted against $1/(2d)$, and $D(N)$ is fitted with Eq. (2).

*Uncertainties.* Each number has four independent contributions, $\sigma^2 = \sigma_{\text{est}}^2 + \sigma_{\text{rip}}^2 + \sigma_{\text{ref}}^2 + \sigma_{\text{res}}^2$: $\sigma_{\text{est}}$, the standard error of the 5 repetitions (negligible); $\sigma_{\text{rip}}$, the ripple of $T$ propagated to the number ($0.92\times$ripple for $D$, and ripple/$|dT/df|$ for an edge); $\sigma_{\text{ref}}$, the spread of the result when it is computed with each reference of the same day; and, for frequencies only, the spectral resolution $\sigma_{\text{res}} = \Delta f/\sqrt{12} = 3.4$ Hz. Per block, $\sigma^2 = \sigma_{\text{disp}}^2 + \sigma_{\text{sys}}^2$. The systematic uncertainty (ripple and reference, common to the whole block) does not decrease with averaging and dominates. Fits are weighted least squares [6]; if $\chi^2/\nu > 1$ the uncertainties are multiplied by $\sqrt{\chi^2/\nu}$. Comparisons use $z = |x - x_0|/\sqrt{\sigma_x^2 + \sigma_0^2}$.

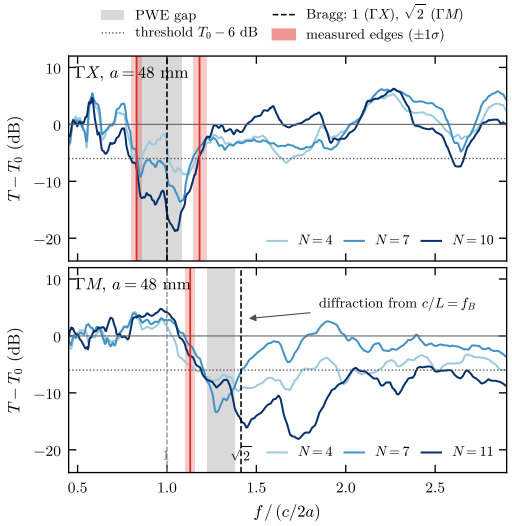



**Figure 2.** Relative transmission for $a = 48$ mm. Grey: PWE gap; red: measured edges (mean over $N \geq 4$). Along $\Gamma X$ the gap closes; along $\Gamma M$ lateral diffraction starts at $f_B$ and $T$ does not return to 0. All $a$ in Appendix A.

## 4 Results

*Gap edges.* Along $\Gamma X$ there is a well-defined gap for $N \geq 4$ for all $a$ (Fig. 2); it closes in 27 of the 28 measurements with $N \geq 4$. The edges are in Table 1: for $a \leq 48$ mm, $f_{\text{inf}}$ agrees with the PWE within $1.3\sigma$ and $f_{\text{sup}}$ within $2.6\sigma$; for $a = 56$ mm both lie above ($z = 3.9$ and 3.6).

**Table 1.** Results per block (mean over $N \geq 4$) versus the PWE. Frequencies in Hz; $\alpha$ in dB/row. † The gap does not close. ‡ Only 14–38 % of the $f_B \pm 10$ % window lies inside the theoretical gap: compares orders of magnitude only.

| | $a$ | $f_B$ | $f_{\text{inf}}$ | | | $f_{\text{sup}}$ | | | $\alpha$ | |
|---|---|---|---|---|---|---|---|---|---|---|
| | mm | | meas. | PWE | $z$ | meas. | PWE | $z$ | meas. | PWE |
| $\Gamma X$ | 32 | 5367 | 3806(84) | 3892 | 1.0 | 5920(190) | 6037 | 0.6 | 0.81(39) | 5.4 |
| | 40 | 4294 | 3590(150) | 3387 | 1.3 | 5120(230) | 4752 | 1.6 | 1.98(41) | 4.0 |
| | 48 | 3578 | 2970(100) | 2988 | 0.2 | 4230(140) | 3880 | 2.6 | 1.45(29) | 2.8 |
| | 56 | 3067 | 2917(62) | 2666 | 3.9 | 3720(120) | 3271 | 3.6 | 1.53(11) | 1.9 |
| $\Gamma M$ | 32 | 7584 | 5500(270) | 6180 | 2.5 | —† | 6990 | — | 0.73(20) | 0.1‡ |
| | 40 | 6067 | 4560(170) | 5109 | 3.1 | 7150(410) | 5818 | 3.2 | 0.81(28) | 0.5‡ |
| | 48 | 5056 | 4040(93) | 4385 | 3.6 | —† | 4942 | — | 1.30(17) | 0.6‡ |

*Dependence on $a$.* The gap does not scale as $1/a$: $f_c/f_B$ rises from 0.90(2) at $a = 32$ mm to 1.09(2) at 56 mm. A line $f_c = m/(2a)$ through the origin gives $m = 339 \pm 15$ m/s, compatible with $c$ ($z = 0.3$) but with $\chi^2/\nu = 15$; the free line ($\chi^2/\nu = 1.4$) gives $m = 229 \pm 20$ m/s and $1280 \pm 230$ Hz.

*Attenuation.* Along $\Gamma X$ the depth grows linearly with $N$ (Fig. 3): for $a = 40$, 48 and 56 mm, the intercept $b = -0.2 \pm 2.0$, $-0.6 \pm 1.9$ and $-0.3 \pm 0.6$ dB which is compatible with 0, as Eq. (2) predicts. For $a = 32$ mm, $D$ saturates at $\approx 13$ dB from $N = 4$, which pulls the fitted line up ($b = 5.1 \pm 2.9$ dB). The measured $\alpha$ is 15, 50, 53 and 80 % of the PWE value for $a = 32$–56 mm.

*Rotated crystal ($\Gamma M$).* Rotating the lattice by 45° moves the gap up, as expected from the closer planes (Fig. 2). The lower edge rises by a factor $f_{\text{inf}}(\Gamma M)/f_{\text{inf}}(\Gamma X) = 1.45 \pm 0.08$, $1.27 \pm 0.07$ and $1.36 \pm 0.06$ for $a = 32$, 40 and 48 mm, consistent with the Bragg estimate $\sqrt{2}$ ($z \leq 2.0$) but below the PWE prediction of 1.59–1.47, because $f_{\text{inf}}(\Gamma M)$ itself lies 8–11 % below the PWE ($z = 2.5$–3.6). Above $f_B$ the gap rarely closes (12 of 24 measurements with $N \geq 4$). The overlap with the $\Gamma X$ gap, $f_{\text{sup}}(\Gamma X) - f_{\text{inf}}(\Gamma M) = 420 \pm 330$, $560 \pm 290$ and $190 \pm 160$ Hz, is below $2\sigma$, while the PWE predicts no overlap ($-140$ to $-500$ Hz).

## 5 Discussion

*Position of the gap along $\Gamma X$.* The theory places the gap below $f_B$ (from $0.93\,f_B$ at $a = 32$ mm to $0.97\,f_B$ at $a = 56$ mm), because the air flowing around the bricks behaves like a slower medium [3]. Our data show a similar trend, but about four

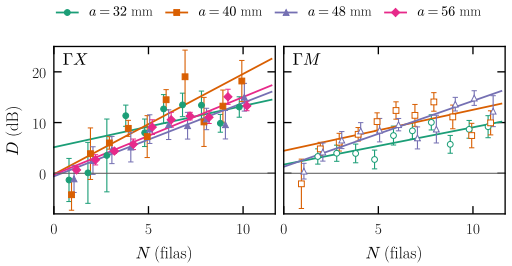



**Figure 3.** Depth $D$ at $f_B \pm 10$ % versus number of rows, with the line $D = \alpha N + b$ fitted for each $a$ (slopes in Table 1).

times stronger. In contrast, for $a = 56$ mm both edges are 9–14 % above the prediction, and we did not find the cause.

*Attenuation along $\Gamma X$.* However, the slope is only 15 to 80 % of the predicted $\alpha$, and $D$ never goes much above 15 dB. We think some sound reaches the microphone without crossing the rows, either through small gaps between the LEGO bricks and the cardboard or around the enclosure. If this sound is about 15 dB weaker than the incident sound, $D$ cannot become larger than that, and a straight line fitted to the data has a smaller slope. The larger the predicted $\alpha$, the sooner $D$ reaches this limit, which agrees with the measured/predicted ratio going from 15 % at $a = 32$ mm to 80 % at $a = 56$ mm.

*Position of the gap along $\Gamma M$.* Above $f_B$, the rows act as a diffraction grating and send part of the sound sideways, which probably explains why the gap does not close. A possible explanation for the low lower edge is that the crystal is only about two wavelengths wide. Because of this, the sound inside it spreads over angles of about $\lambda/W \approx 30°$, and the part that travels closer to $\Gamma X$ may already be blocked by the $\Gamma X$ gap: the measured $f_{\text{inf}}$ along $\Gamma M$ (1.02–1.13 times $f_B$ along $\Gamma X$) lies inside the $\Gamma X$ gap for the same $a$. However, our data cannot confirm this.

*Attenuation along $\Gamma M$.* Along $\Gamma M$ the side walls were made of foam instead of LEGO, so sound could go around the crystal more easily, which makes the gap shallower. The comparison with the theory is rough here: the theoretical gap covers only 14–38 % of the $f_B \pm 10$ % window where $D$ is measured, and the theory predicts almost no attenuation there. Therefore, the measured attenuation must come mostly from other effects, such as sound crossing the crystal at an angle.

*Finite crystal.* The theory we used is for an infinite crystal. The gap needs a few rows to form: for $N \leq 2$ we almost never see it, so we only average $N \geq 4$. Along $\Gamma X$ the LEGO walls reflect the sound like mirrors, so the 4 columns behave roughly like a wider crystal. Along $\Gamma M$ the foam does not reflect the sound, so the crystal is far from infinite.

*Limitations.* The lattice constant $a$ can only change in steps of 8 mm, and the columns are only 3.8 cm tall and not perfectly smooth or sealed. Background noise may also be a problem: the measurements were usually made with loud noise from other experiments in the lab (this effect is reduced because there were 5 recordings for each configuration). As mentioned in the analysis, the largest source of error is the reference.

## 6 Conclusion

A LEGO crystal has a Bragg gap along $\Gamma X$ whose lower edge agrees with the PWE within $1.3\sigma$ for $a \leq 48$ mm, and which does not scale as $1/a$. The attenuation is linear in $N$, limited to $\approx 15$ dB by sound that goes around the crystal. Along $\Gamma M$ the gap appears about $\sqrt{2}$ higher, but the lower edge is 8–11 % below the PWE and the gap does not close; there is no evidence of a complete gap.

## References


[1] R. Martínez-Sala *et al.*, Nature **378**, 241 (1995).
[2] J. V. Sánchez-Pérez *et al.*, Phys. Rev. Lett. **80**, 5325 (1998).
[3] A. A. Krokhin *et al.*, Phys. Rev. Lett. **91**, 264302 (2003).
[4] M. S. Kushwaha *et al.*, Phys. Rev. Lett. **71**, 2022 (1993).
[5] P. D. Welch, IEEE Trans. Audio Electroacoust. **15**, 70 (1967).
[6] P. R. Bevington, D. K. Robinson, *Data Reduction and Error Analysis for the Physical Sciences*, McGraw-Hill (2003).

# A Additional data and measurements

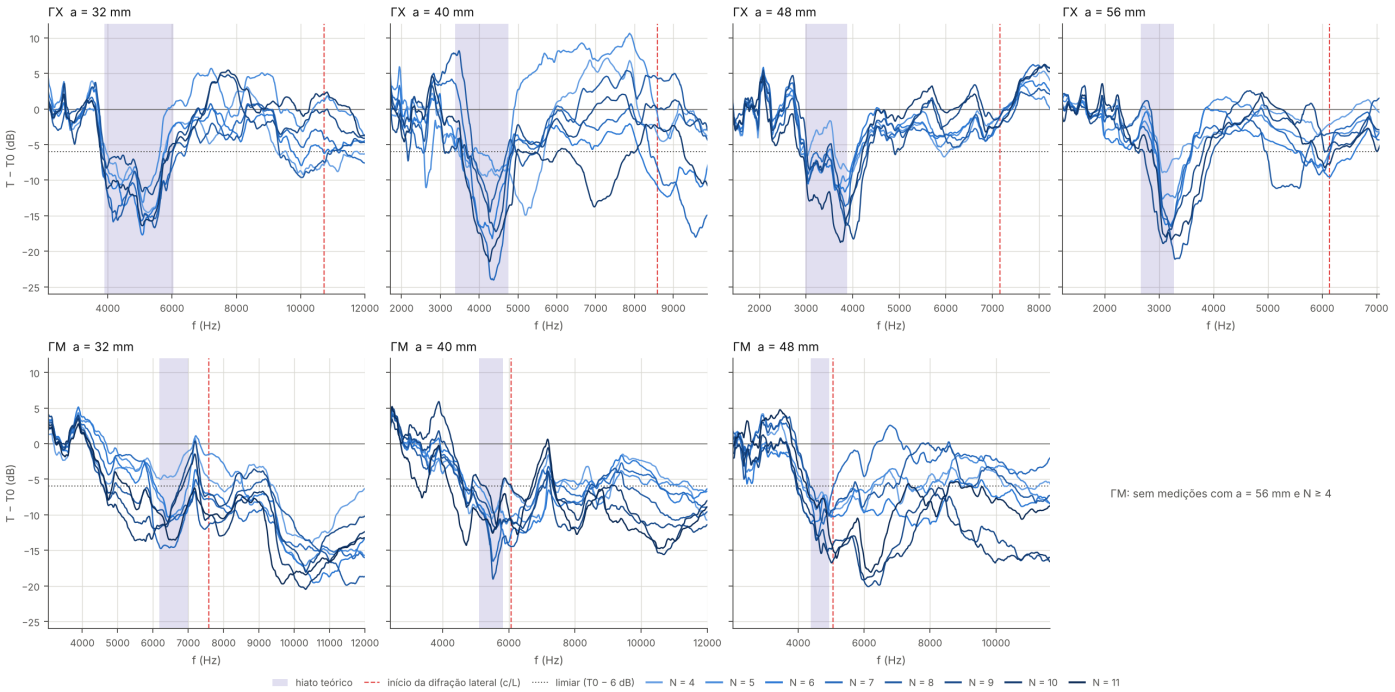



**Figure 4.** $T - T_0$ for all blocks with $N \geq 4$ (top: $\Gamma X$; bottom: $\Gamma M$). Band: PWE gap; red dashed: onset of lateral diffraction, $c/L$; dotted: threshold $T_0 - 6\,\mathrm{dB}$. Along $\Gamma M$ there are no measurements with $a = 56\,\mathrm{mm}$ and $N \geq 4$.

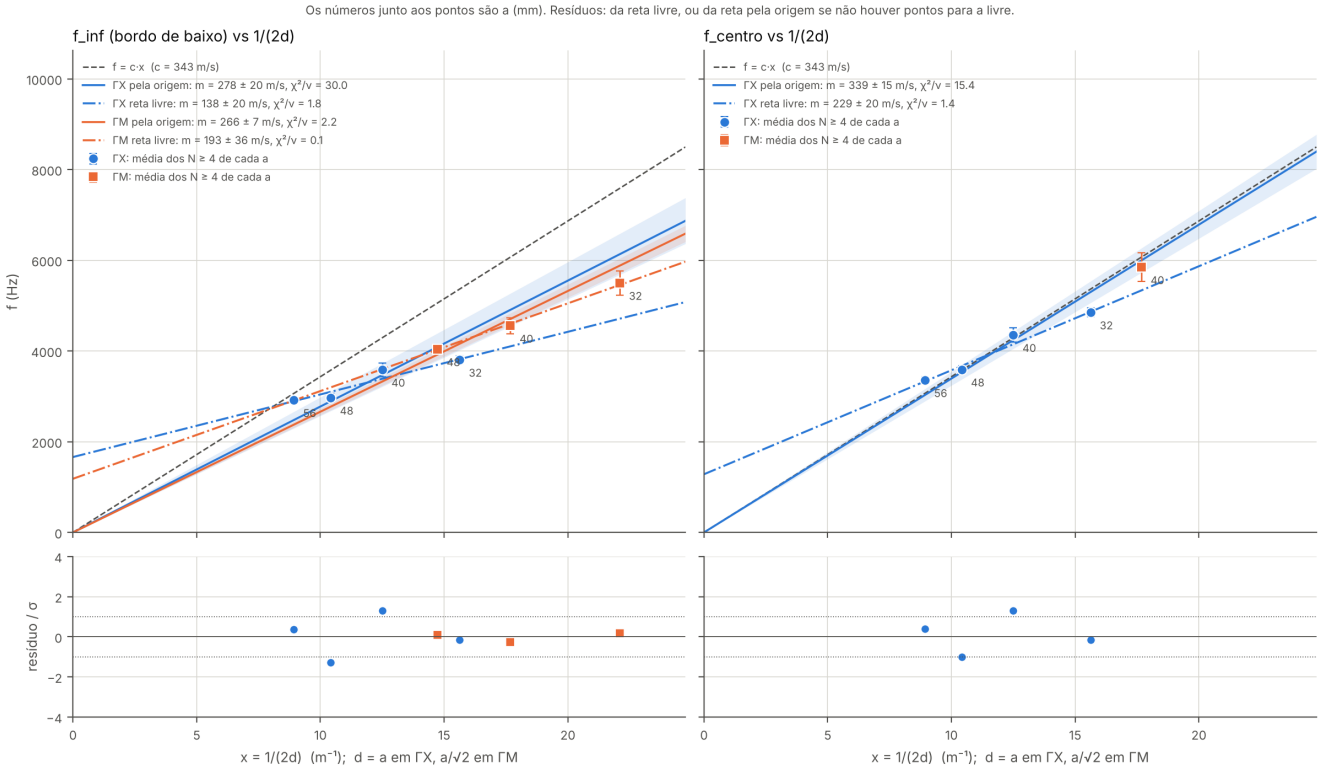



**Figure 5.** $f_{\mathrm{inf}}$ and $f_c$ (mean over $N \geq 4$ for each $a$) versus $x = 1/(2d)$, with the line through the origin and the free line; below, normalised residuals. Dashed: $f = c\,x$.

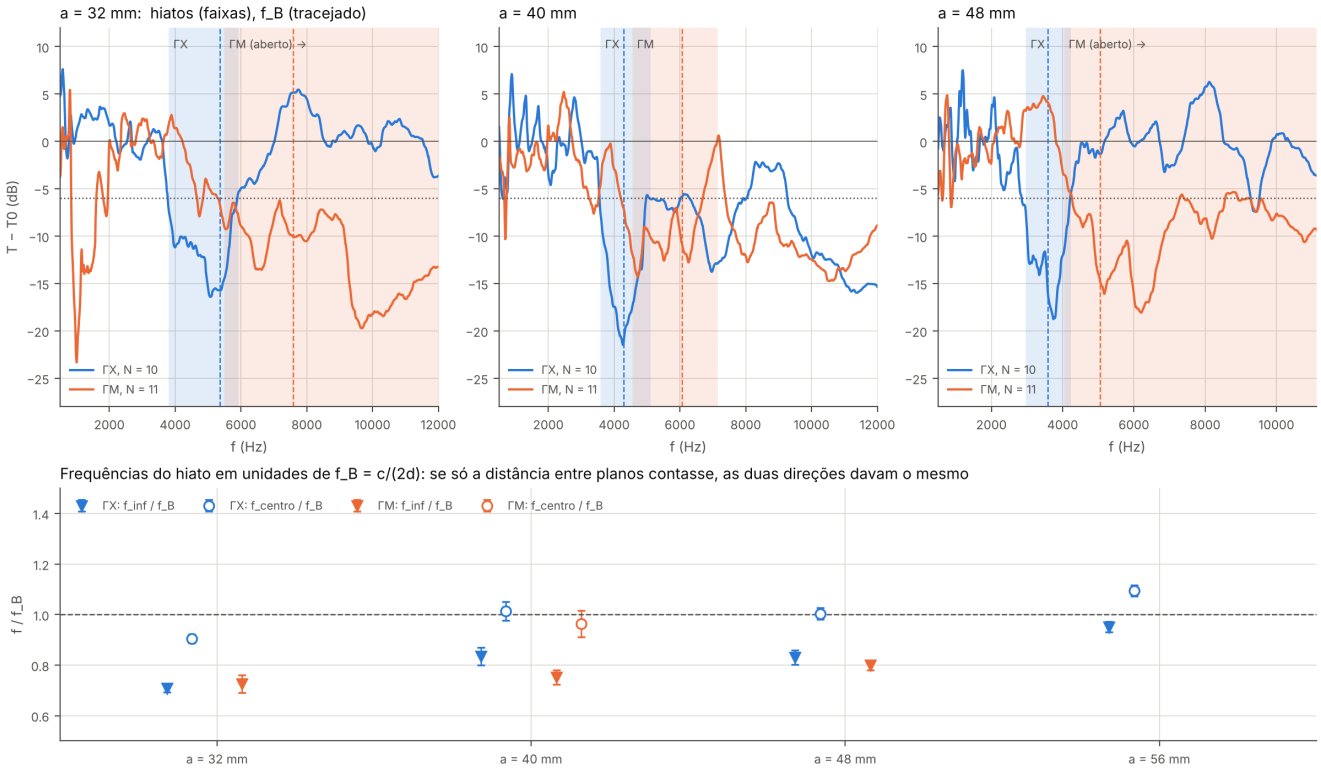



**Figure 6.** Top: $\Gamma X$ ($N = 10$) and $\Gamma M$ ($N = 11$) for the same $a$, with the measured gaps (bands) and $f_B$ (dashed). Bottom: $f_{\mathrm{inf}}/f_B$ and $f_c/f_B$ along both directions.

**Table 2.** All measurements. "rip.": ripple of $T$ in the control band (dB); "$k/5$ rep.": repetitions used; "ref. other $a$": reference from the same session but from another block. Total uncertainty $\sigma^2 = \sigma^2_{\text{est}} + \sigma^2_{\text{rip}} + \sigma^2_{\text{ref}}$ ($+\sigma^2_{\text{res}}$). The components are also in `tabela.csv`.

| dir | $a$ (mm) | $N$ | day | reference | rip. | $D$ (dB) | $f_{\text{inf}}$ (Hz) | $f_{\text{sup}}$ (Hz) | notes |
|---|---|---|---|---|---|---|---|---|---|
| $\Gamma X$ | 32 | 1 | 10-01 | ref_a32 | 4.2 | $-1.3 \pm 4.3$ | — | — | |
| $\Gamma X$ | 32 | 2 | 10-01 | ref_a32 | 6.2 | $0.1 \pm 6.0$ | — | — | |
| $\Gamma X$ | 32 | 3 | 10-01 | ref_a32 | 7.6 | $3.5 \pm 7.2$ | $5529 \pm 889$ | $5609 \pm 599$ | |
| $\Gamma X$ | 32 | 4 | 10-01 | ref_a32 | 1.7 | $11.4 \pm 2.1$ | $3947 \pm 195$ | $5768 \pm 1747$ | 4/5 rep. |
| $\Gamma X$ | 32 | 5 | 10-01 | ref_a32 | 1.8 | $8.0 \pm 2.7$ | $3846 \pm 133$ | $5622 \pm 161$ | |
| $\Gamma X$ | 32 | 6 | 10-01 | ref_a32 | 2.2 | $12.7 \pm 2.8$ | $3795 \pm 67$ | $5802 \pm 109$ | |
| $\Gamma X$ | 32 | 7 | 10-01 | ref_a32 | 1.7 | $13.5 \pm 2.3$ | $3781 \pm 70$ | $6431 \pm 165$ | |
| $\Gamma X$ | 32 | 8 | 10-01 | ref_a32 | 2.1 | $13.4 \pm 2.8$ | $3880 \pm 91$ | $6028 \pm 181$ | |
| $\Gamma X$ | 32 | 9 | 10-01 | ref_a32 | 1.5 | $9.9 \pm 1.8$ | $3794 \pm 79$ | $5981 \pm 893$ | |
| $\Gamma X$ | 32 | 10 | 10-01 | ref_a32 | 1.6 | $13.1 \pm 2.0$ | $3773 \pm 78$ | $5890 \pm 919$ | 4/5 rep. |
| $\Gamma X$ | 40 | 1 | 10-06 | ref_a56 | 3.1 | $-4.2 \pm 3.1$ | — | — | 4/5 rep., ref. other $a$ |
| $\Gamma X$ | 40 | 2 | 10-06 | ref_a56 | 5.4 | $3.9 \pm 5.1$ | $4538 \pm 944$ | $4713 \pm 254$ | ref. other $a$ |
| $\Gamma X$ | 40 | 3 | 10-06 | ref_a56 | 1.0 | $6.0 \pm 1.2$ | $3196 \pm 330$ | $4832 \pm 126$ | 4/5 rep., ref. other $a$ |
| $\Gamma X$ | 40 | 4 | 10-06 | ref_a56 | 1.2 | $8.9 \pm 1.6$ | $3474 \pm 168$ | $5677 \pm 114$ | ref. other $a$ |
| $\Gamma X$ | 40 | 5 | 10-06 | ref_a56 | 4.2 | $7.2 \pm 4.1$ | $3625 \pm 327$ | $4783 \pm 163$ | ref. other $a$ |
| $\Gamma X$ | 40 | 6 | 10-06 | ref_a56 | 2.0 | $14.5 \pm 2.0$ | $3525 \pm 122$ | $4818 \pm 271$ | ref. other $a$ |
| $\Gamma X$ | 40 | 7 | 10-01 | ref_a40 | 4.4 | $19.0 \pm 5.3$ | $3696 \pm 157$ | $4875 \pm 185$ | |
| $\Gamma X$ | 40 | 8 | 10-01 | ref_a40 | 4.7 | $10.1 \pm 4.8$ | $3833 \pm 213$ | $5510 \pm 462$ | |
| $\Gamma X$ | 40 | 9 | 10-01 | ref_a40 | 1.9 | $13.2 \pm 3.2$ | $3669 \pm 330$ | $4854 \pm 111$ | |
| $\Gamma X$ | 40 | 10 | 10-01 | ref_a40 | 2.2 | $18.2 \pm 4.1$ | $3556 \pm 101$ | — | does not close |
| $\Gamma X$ | 48 | 1 | 10-06 | ref_a48 | 1.9 | $-1.1 \pm 2.7$ | — | — | |
| $\Gamma X$ | 48 | 2 | 10-06 | ref_a48 | 2.4 | $2.5 \pm 2.7$ | — | — | 4/5 rep. |
| $\Gamma X$ | 48 | 3 | 10-06 | ref_a48 | 2.3 | $4.6 \pm 2.7$ | $3600 \pm 502$ | $3770 \pm 1598$ | |
| $\Gamma X$ | 48 | 4 | 10-06 | ref_a48 | 2.3 | $5.2 \pm 2.9$ | $3613 \pm 568$ | $4173 \pm 364$ | |
| $\Gamma X$ | 48 | 5 | 10-06 | ref_a48 | 2.7 | $8.7 \pm 2.8$ | $2883 \pm 125$ | $4157 \pm 260$ | |
| $\Gamma X$ | 48 | 6 | 10-06 | ref_a48 | 2.5 | $9.6 \pm 2.9$ | $2970 \pm 100$ | $4071 \pm 161$ | 4/5 rep. |
| $\Gamma X$ | 48 | 7 | 10-06 | ref_a48 | 2.0 | $9.4 \pm 2.6$ | $2982 \pm 98$ | $4093 \pm 130$ | 4/5 rep. |
| $\Gamma X$ | 48 | 8 | 10-06 | ref_a48 | 2.0 | $10.7 \pm 2.1$ | $2974 \pm 83$ | $4306 \pm 112$ | |
| $\Gamma X$ | 48 | 9 | 10-06 | ref_a48 | 2.3 | $9.6 \pm 2.8$ | $3019 \pm 79$ | $4332 \pm 96$ | |
| $\Gamma X$ | 48 | 10 | 10-06 | ref_a48 | 2.8 | $15.0 \pm 2.7$ | $2850 \pm 135$ | $4210 \pm 123$ | |
| $\Gamma X$ | 56 | 1 | 10-06 | ref_a56 | 0.8 | $0.7 \pm 0.7$ | — | — | |
| $\Gamma X$ | 56 | 2 | 10-06 | ref_a56 | 1.1 | $2.7 \pm 1.0$ | — | — | 4/5 rep. |
| $\Gamma X$ | 56 | 3 | 10-06 | ref_a56 | 0.5 | $4.4 \pm 0.5$ | $3018 \pm 50$ | $3142 \pm 230$ | |
| $\Gamma X$ | 56 | 4 | 10-06 | ref_a56 | 1.1 | $5.7 \pm 1.0$ | $2993 \pm 75$ | $3418 \pm 200$ | |
| $\Gamma X$ | 56 | 5 | 10-06 | ref_a56 | 1.6 | $9.1 \pm 1.5$ | $2912 \pm 72$ | $3607 \pm 133$ | 4/5 rep. |
| $\Gamma X$ | 56 | 6 | 10-06 | ref_a56 | 1.6 | $10.6 \pm 1.5$ | $2941 \pm 56$ | $3587 \pm 98$ | 4/5 rep. |
| $\Gamma X$ | 56 | 7 | 10-06 | ref_a56 | 0.8 | $11.3 \pm 0.7$ | $2884 \pm 46$ | $3593 \pm 77$ | |
| $\Gamma X$ | 56 | 8 | 10-06 | ref_a56 | 1.3 | $11.0 \pm 1.2$ | $2979 \pm 47$ | $3993 \pm 103$ | |
| $\Gamma X$ | 56 | 9 | 10-06 | ref_a56 | 1.5 | $15.1 \pm 1.5$ | $2595 \pm 266$ | $3809 \pm 215$ | |
| $\Gamma X$ | 56 | 10 | 10-06 | ref_a56 | 1.0 | $13.3 \pm 0.9$ | $2821 \pm 58$ | $3886 \pm 85$ | |
| $\Gamma M$ | 32 | 2 | 10-13 | ref_a32 | 1.6 | $3.3 \pm 1.5$ | $6288 \pm 297$ | $6883 \pm 150$ | |
| $\Gamma M$ | 32 | 3 | 10-13 | ref_a32 | 1.7 | $4.0 \pm 1.6$ | $5950 \pm 96$ | $6953 \pm 128$ | |
| $\Gamma M$ | 32 | 4 | 10-13 | ref_a32 | 2.0 | $3.9 \pm 1.8$ | $7871 \pm 1018$ | $8223 \pm 817$ | 4/5 rep. |
| $\Gamma M$ | 32 | 5 | 10-13 | ref_a32 | 2.0 | $2.7 \pm 1.9$ | $6005 \pm 2114$ | $6989 \pm 120$ | |
| $\Gamma M$ | 32 | 6 | 10-13 | ref_a32 | 2.0 | $7.4 \pm 1.9$ | $6076 \pm 179$ | — | does not close |
| $\Gamma M$ | 32 | 7 | 10-13 | ref_a32 | 1.8 | $8.4 \pm 1.6$ | $5910 \pm 73$ | — | 4/5 rep., does not close |
| $\Gamma M$ | 32 | 8 | 10-13 | ref_a32 | 1.6 | $10.0 \pm 1.5$ | $4603 \pm 189$ | — | does not close |
| $\Gamma M$ | 32 | 9 | 10-13 | ref_a32 | 1.9 | $5.7 \pm 1.8$ | $4680 \pm 206$ | $8281 \pm 309$ | |
| $\Gamma M$ | 32 | 10 | 10-13 | ref_a32 | 1.9 | $8.7 \pm 1.8$ | $4581 \pm 141$ | — | does not close |
| $\Gamma M$ | 32 | 11 | 10-13 | ref_a32 | 2.4 | $9.2 \pm 2.2$ | $4664 \pm 353$ | — | does not close |
| $\Gamma M$ | 40 | 1 | 10-13 | ref_a40_fim | 5.3 | $-2.1 \pm 4.9$ | — | — | |
| $\Gamma M$ | 40 | 2 | 10-13 | ref_a40_fim | 1.4 | $4.8 \pm 1.3$ | $6260 \pm 302$ | $6794 \pm 1291$ | |
| $\Gamma M$ | 40 | 3 | 10-13 | ref_a40_fim | 1.9 | $6.1 \pm 1.7$ | $4712 \pm 335$ | $5756 \pm 593$ | 4/5 rep. |
| $\Gamma M$ | 40 | 4 | 10-13 | ref_a40_fim | 1.4 | $7.7 \pm 1.3$ | $4633 \pm 101$ | — | 4/5 rep., does not close |
| $\Gamma M$ | 40 | 5 | 10-13 | ref_a40_fim | 1.9 | $10.2 \pm 1.7$ | $4695 \pm 445$ | $7151 \pm 209$ | |
| $\Gamma M$ | 40 | 6 | 10-13 | ref_a40_fim | 1.9 | $12.4 \pm 1.7$ | $4706 \pm 213$ | $8699 \pm 493$ | |
| $\Gamma M$ | 40 | 7 | 10-13 | ref_a40_fim | 2.3 | $11.4 \pm 2.2$ | $4535 \pm 210$ | $6865 \pm 1507$ | |
| $\Gamma M$ | 40 | 8 | 10-13 | ref_a40_fim | 2.0 | $14.1 \pm 1.8$ | $4544 \pm 175$ | $8655 \pm 429$ | |
| $\Gamma M$ | 40 | 9 | 10-13 | ref_a40_fim | 2.5 | $11.1 \pm 2.3$ | $4624 \pm 342$ | — | does not close |
| $\Gamma M$ | 40 | 10 | 10-13 | ref_a40_fim | 2.6 | $7.4 \pm 2.4$ | $4616 \pm 184$ | — | does not close |
| $\Gamma M$ | 40 | 11 | 10-13 | ref_a40_fim | 2.6 | $9.9 \pm 2.4$ | $4206 \pm 165$ | $6696 \pm 177$ | 4/5 rep. |
| $\Gamma M$ | 48 | 1 | 10-12 | ref_a48_fim | 1.4 | $0.4 \pm 1.5$ | — | — | |
| $\Gamma M$ | 48 | 2 | 10-12 | ref_a48_fim | 1.3 | $4.2 \pm 1.8$ | $5838 \pm 656$ | — | 4/5 rep. |
| $\Gamma M$ | 48 | 3 | 10-12 | ref_a48_fim | 1.2 | $6.5 \pm 1.7$ | $3896 \pm 112$ | — | |
| $\Gamma M$ | 48 | 4 | 10-12 | ref_a48_fim | 1.3 | $8.3 \pm 1.6$ | $4148 \pm 215$ | $6287 \pm 557$ | |
| $\Gamma M$ | 48 | 5 | 10-12 | ref_a48_fim | 1.9 | $8.5 \pm 2.3$ | $4131 \pm 176$ | $6232 \pm 769$ | |
| $\Gamma M$ | 48 | 6 | 10-12 | ref_a48_fim | 1.0 | $9.6 \pm 1.2$ | $3966 \pm 61$ | — | does not close |
| $\Gamma M$ | 48 | 7 | 10-12 | ref_a48_fim | 1.7 | $7.0 \pm 2.2$ | $4314 \pm 76$ | $5068 \pm 144$ | |
| $\Gamma M$ | 48 | 8 | 10-12 | ref_a48_fim | 2.7 | $8.7 \pm 2.7$ | $4360 \pm 236$ | $6941 \pm 973$ | |
| $\Gamma M$ | 48 | 9 | 10-12 | ref_a48_fim | 0.9 | $13.6 \pm 1.3$ | $3945 \pm 50$ | — | does not close |
| $\Gamma M$ | 48 | 10 | 10-12 | ref_a48_fim | 1.2 | $14.8 \pm 1.4$ | $3990 \pm 85$ | — | does not close |
| $\Gamma M$ | 48 | 11 | 10-12 | ref_a48_fim | 2.1 | $12.2 \pm 3.0$ | $4260 \pm 188$ | — | does not close |

## B Calibrations and derivations

### Speed of sound

$c(\theta) = 331.3\sqrt{1+\theta/273.15}\,\mathrm{m/s}$, $\sigma_c = c\,\sigma_\theta/[2(273.15+\theta)] = 0.6\,\mathrm{m/s}$ for $\sigma_\theta = 1\,^\circ\mathrm{C}$.

### Plane-wave Criterion

The loudspeaker emits approximately spherical waves. For the wave at the crystal entrance to be treated as plane, the distance $r$ from the loudspeaker to the entrance must be large compared with the entrance width $W$. The loudspeaker faces the centre of the entrance, so the entrance edges are $W/2$ from the axis and are reached after an extra path, relative to the centre, of $\delta = \sqrt{r^2 + (W/2)^2} - r \approx W^2/(8r)$. This extra path corresponds to a phase difference of $360^\circ\,\delta/\lambda$ across the entrance, and we require $\delta \le \lambda/10$ (at most $36^\circ$), which gives

$$r \ge \frac{1.25\,W^2}{\lambda_{\min}}, \tag{4}$$

where $\lambda_{\min} = c/f_{\max}$ is the smallest wavelength used in the analysis. Since $r \propto \lambda^{-1}$, the higher the frequency, the larger the necessary distance $r$. The highest frequency we use is the upper gap edge, which in Table 1 is at most $1.21\,f_B$ ($a = 56\,\mathrm{mm}$), so we take $f_{\max} = 1.2 f_B$, i.e. $\lambda_{\min} = 2d/1.2$. Along $\Gamma X$, $W = 18$ studs $= 144\,\mathrm{mm}$ for all $a$; along $\Gamma M$, $W = 14$, 17 and 20 studs (112, 136 and 160 mm) for $a = 32$, 40 and 48 mm.

**Table 3.** Minimum distance $r_{\min} = 1.25\,W^2/\lambda_{\min}$, with $\lambda_{\min} = 2d/1.2$ (upper edge of the gap), and margin with respect to $r = 70\,\mathrm{cm}$.

| Lattice | $a$ (mm) | $d$ (mm) | $W$ (mm) | $\lambda_{\min}$ (mm) | $r_{\min}$ (cm) | $70\,\mathrm{cm}/r_{\min}$ |
|---|---|---|---|---|---|---|
| $\Gamma X$ | 32 | 32.0 | 144 | 53.3 | 48.6 | 1.44 |
| $\Gamma X$ | 40 | 40.0 | 144 | 66.7 | 38.9 | 1.80 |
| $\Gamma X$ | 48 | 48.0 | 144 | 80.0 | 32.4 | 2.16 |
| $\Gamma X$ | 56 | 56.0 | 144 | 93.3 | 27.8 | 2.52 |
| $\Gamma M$ | 32 | 22.6 | 112 | 37.7 | 41.6 | 1.68 |
| $\Gamma M$ | 40 | 28.3 | 136 | 47.1 | 49.0 | 1.43 |
| $\Gamma M$ | 48 | 33.9 | 160 | 56.6 | 56.6 | 1.24 |

In all configurations $r = 70\,\mathrm{cm}$ exceeds $r_{\min}$, with a minimum margin of 1.24 ($\Gamma M$, $a = 48\,\mathrm{mm}$).

### Plane-wave expansion (PWE)

With $\sigma = 1/\rho$, $\eta = 1/(\rho c^2)$ and $p(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\sum_{\mathbf{G}} p_{\mathbf{G}}\, e^{i\mathbf{G}\cdot\mathbf{r}}$, $\mathbf{G} = \frac{2\pi}{a}(m,n)$, the wave equation gives, for each $\mathbf{k}$, the eigenvalue problem

$$\sum_{\mathbf{G}'} \sigma(\mathbf{G}-\mathbf{G}')\,(\mathbf{k}+\mathbf{G})\cdot(\mathbf{k}+\mathbf{G}')\,p_{\mathbf{G}'} = \omega^2 \sum_{\mathbf{G}'} \eta(\mathbf{G}-\mathbf{G}')\,p_{\mathbf{G}'},$$

with a square rod of side $s$, $\sigma(\mathbf{G}) = (\sigma_{\mathrm{t}} - \sigma_{\mathrm{air}})\,ff\,\mathrm{sinc}(G_x s/2)\,\mathrm{sinc}(G_y s/2)$ ($\mathbf{G} \neq 0$). We use $15^2 = 225$ waves (729 waves change the edges by 0.1–0.4 %). Only modes symmetric with respect to the plane containing the sound direction count (the antisymmetric ones are not excited by a wave arriving head-on). The gap is $f_{\mathrm{inf}} = \max_k f_1(k)$, $f_{\mathrm{sup}} = \min_k f_{n\ge 2}(k)$, with $k$ from $\Gamma$ to X ($|\mathbf{k}| = \pi/a$) or from $\Gamma$ to M ($|\mathbf{k}| = \sqrt{2}\pi/a$). The attenuation comes from the inverse problem (given $\omega$, a complex $k$ comes out), $(k^2 M_2 + k M_1 + M_0)\,\mathbf{p} = 0$, linearised into a problem of double dimension; $\kappa$ is the smallest $|\,\mathrm{Im}\,k|$. With only the two waves $e^{\pm i\pi x/d}$, $f_{\mathrm{sup}} - f_{\mathrm{inf}} \propto |\sigma(2\pi/d)|$: the gap widens with the amount of brick per plane.

### Effective medium

At low frequency, $\kappa_{\mathrm{eff}} = (1-ff)\kappa$ and $\rho_{\mathrm{eff}} \simeq \rho(1+ff)/(1-ff)$ (rigid rods), hence $c_{\mathrm{eff}} = c/\sqrt{1+ff}$ and $f_{\mathrm{gap}} \approx f_B/\sqrt{1+ff} = 0.90$, 0.93, 0.95, $0.96\,f_B$ for $a = 32$–$56\,\mathrm{mm}$.

### Complete gap

A complete band gap requires the upper edge of the first band to lie below the lower edge of the next band for every wave vector in the irreducible Brillouin zone, $\max_{\mathbf{k}} f_1(\mathbf{k}) < \min_{\mathbf{k}} f_2(\mathbf{k})$. In practice, the limiting directions are expected to be close to $\Gamma X$ and $\Gamma M$, so we compare the stop bands obtained along these two directions. A necessary overlap condition is $f_{\mathrm{sup}}(\Gamma X) > f_{\mathrm{inf}}(\Gamma M)$. Along $\Gamma M$ the diagonal of the brick, 22.3 mm, occupies almost the whole distance between adjacent planes, since $\frac{a}{\sqrt{2}} = 22.6\,\mathrm{mm}$ for $a = 32\,\mathrm{mm}$. The medium therefore appears more nearly uniform along this direction and the $\Gamma M$ stop band is narrow. The predicted ratios $\frac{f_{\mathrm{inf}}(\Gamma M)}{f_{\mathrm{inf}}(\Gamma X)}$ are 1.59, 1.51 and 1.47, rather than exactly $\sqrt{2}$. For the $2\times 2$ bricks considered here, the $\Gamma X$ and $\Gamma M$ stop bands do not overlap, with a remaining separation of about 7 Hz even for $a = 24\,\mathrm{mm}$. Thus the calculations do not support a complete band gap for these configurations.

### Uncertainty of the depth

$D = T_0 - M$; with $n_{\mathrm{base}} = \log_2(0.70/0.35)/(1/6) = 6$ and $n_{\mathrm{win}} = \log_2(1.1/0.9)/(1/6) = 1.74$ independent points, and the variance of the median being $\pi/2$ times that of the mean, $\mathrm{Var}\,D = \left(\frac{\pi}{2}\frac{1}{6} + \frac{1}{1.74}\right)\sigma^2$, i.e. $\sigma_{\mathrm{rip}}(D) = 0.92 \times$ ripple. The dispersion of the references uses weights $w_k = 1/\mathrm{rip}_k^2$: $\sigma_{\mathrm{ref}}^2 = \sum_k w_k (x_k - x_0)^2 / \sum_k w_k$.

### Fits

For $y = mx + b$ with $w_i = 1/\sigma_i^2$: $m = (S\,S_{xy} - S_x S_y)/\Delta$, $b = (S_{xx} S_y - S_x S_{xy})/\Delta$, $\sigma_m^2 = S/\Delta$, $\sigma_b^2 = S_{xx}/\Delta$, $\Delta = S\,S_{xx} - S_x^2$ [6], the uncertainty in $x$ enters through $\sigma_{i,\mathrm{eff}}^2 = \sigma_{y,i}^2 + (m\sigma_{x,i})^2$, minimising $\chi^2(m,b)$ numerically. If $\chi^2/\nu > 1$, $\sigma \to \sqrt{\chi^2/\nu}\,\sigma$, uncertainties are never shrunk. The regressions were checked against `numpy.polyfit`, `scipy.optimize.curve_fit` and `scipy.odr`.

## C Code

Repository: [link to be added].

- `cristal_simples/` and `cristal_lego_g/cristal_lego_g/`: measurement apps ($\Gamma X$ and $\Gamma M$; Python, `sounddevice`) with the measurement plan, quality control and the recordings (`projeto/dados/<step>/*.wav` + `meta.json`; repeated measurements in `_antigos/`, which do not count).
- `cristal_lego_g/analise 2.0/`: joint analysis of the two directions. `analise.py` (spectra, $T(f)$, quantities, uncertainties, fits), `teoria.py` (PWE), `graficos.py`, `app.py` (interface); results in `resultados/` (`resultados.txt`, `tabela.csv`, `ajustes.csv`, `teoria.txt`). All equations, in order, in `EQUACOES.md`.
- `relatorio/`: this report. `fazer_figuras.py` generates Figs. 2–3 and Table A1 from `dados_figuras/`; `dados_figuras/curvas_T.py` recomputes the curves of Fig. 2 from the `.wav` files (same method, numpy only).

Parameters: FS = 48000, NPERSEG = 4096, band 0.5–12 kHz, segment 0.8–10.3 µs, 1/6-octave smoothing, 6 dB threshold, $\sigma_a = 0.1$ mm, $\sigma_\theta = 1\,^\circ$C.

## D Use of AI

We used Claude (Anthropic) as an assistant: in writing the measurement and analysis code (which we reviewed and tested: the regressions were compared with independent libraries and the coverage of the uncertainties by simulation) and in drafting this report from the results files. The set-up, the measurements, the decisions on what to measure and the verification of the numbers against `resultados.txt` and `teoria.txt` were done by us.